Supplementary Information

Benchmarking the accuracy, robustness and reproducibility of spatial domain identification methods

Supplementary Notes

Supplementary Note 1. Data and preprocessing

DLPFC count matrices from spatialLIBD and layer annotations from the SEDR analyses repository; STARmap and MERFISH matrices and annotations from the BASS analyses. Genes in fewer than 50 spots or with fewer than 10 counts were removed for Visium; 2000 highly variable genes were selected on the counts (Seurat v3 flavour; Visium) or all panel genes kept, then counts were normalised to the library size, log-transformed, scaled and reduced by PCA (200 components for Visium, 50 for panels). Sections, splits and label provenance are listed in configs/datasets.yaml; per-section statistics are in Supplementary Table S1.

Supplementary Note 2. Methods and settings

Every method was run from its official code with the settings of its tutorial or paper (Supplementary Table S2). The true number of domains was given to every method. Neighbourhood radii given in pixels were converted to multiples of the spot or cell spacing so that they transfer across technologies. Methods that output an embedding were additionally clustered with matched clustering: a 20-start tied-covariance Gaussian mixture keeping the best likelihood, the same for every method.

Supplementary Note 3. Metrics

Agreement: ARI, NMI, AMI. Per-domain: clusters matched to domains with the Hungarian algorithm; macro-F1, missed domains, recall of domains below 5% of spots. Spatial: CHAOS (mean distance to the nearest same-cluster neighbour), PAS (fraction of spots whose cluster differs from most of their ten neighbours), Moran's I of the labels, boundary F1. Label-free: silhouette, Davies–Bouldin, Calinski–Harabasz. Definitions and code: spatialbench/metrics.py (unit tests in tests/).

Supplementary Note 4. Statistics

Sections are nested in groups (three DLPFC donors, STARmap, MERFISH). Differences to the reference are equal-weight means of group means with 95% percentile intervals from a hierarchical bootstrap (5000 resamples: groups, then sections within groups). Equivalence: 90% interval within ± 0.03 ARI, about the seed-to-seed standard deviation. No correction for multiple comparisons is applied. Ranks and the probability of ranking first come from the same resamples. The reference method was chosen after seeing the means, so these comparisons are exploratory. Code: `scripts/stats_cluster_aware.py`.

Supplementary Note 5. Audit of released code

Repositories were screened with `scripts/audit_leakage.py` for five patterns of label use (Supplementary Table S18), and every candidate line was read by one author. The taxonomy and all judgements, including false alarms, are in `docs/LEAKAGE_AUDIT.md`. The audit describes what the released code does; it does not establish how published results were obtained.

Supplementary Note 6. Computing environment

All runs used one CPU thread per job on a 4-core Intel Xeon (2.8 GHz) with 15 GB of memory, Python 3.13, R 4.3.3 and mclust 6.0.1. Memory-heavy methods (DeepST, BayesSpace, NichePCA) were run one job at a time.

Supplementary Tables

Supplementary Table S1. Data and quality control per section. Sparsity: fraction of zero entries. Annotated: fraction of spots or cells with a domain label. Smallest domain: fraction of annotated spots in the smallest domain. Imbalance: largest over smallest domain size.

Section	Spots/cells	Genes	Median counts	Sparsity	Annotated	Domains	Smallest domain	Imbalance
DLPFC 151507	4226	33538	2464	0.96	1.00	7	0.072	4.0
DLPFC 151508	4384	33538	2019	0.96	1.00	7	0.046	6.9
DLPFC 151509	4789	33538	2416	0.96	1.00	7	0.035	11.3
DLPFC 151510	4634	33538	2319	0.96	0.99	7	0.039	9.9
DLPFC 151669	3661	33538	3470	0.95	0.99	5	0.063	9.3
DLPFC 151670	3498	33538	3113	0.95	1.00	5	0.060	10.4
DLPFC 151671	4110	33538	3743	0.94	1.00	5	0.060	7.8
DLPFC 151672	4015	33538	3482	0.95	0.97	5	0.078	5.2
DLPFC 151673	3639	33538	4120	0.93	0.99	7	0.060	4.5
DLPFC 151674	3673	33538	5334	0.92	0.99	7	0.062	4.1
DLPFC 151675	3592	33538	3140	0.95	0.99	7	0.077	2.8
DLPFC 151676	3460	33538	3419	0.94	0.99	7	0.074	3.3
STARmap 20180417_BZ5_control	1049	166	604	0.69	1.00	4	0.082	5.2
STARmap 20180419_BZ9_control	1053	166	301	0.77	1.00	4	0.078	6.1
STARmap 20180424_BZ14_control	1088	166	393	0.70	1.00	4	0.031	13.6
MERFISH -0.04	5488	155	241	0.58	1.00	8	0.019	13.1
MERFISH -0.09	5557	155	260	0.57	1.00	8	0.025	9.7
MERFISH -0.14	5926	155	246	0.57	1.00	8	0.036	7.8
MERFISH -0.19	5803	155	254	0.58	1.00	8	0.043	5.6
MERFISH -0.24	5543	155	244	0.59	1.00	8	0.041	7.0

Supplementary Table S2. Benchmarked methods: code source, version, settings source, clustering and every deviation from the published tutorial (from configs/methods.yaml).

Method	Type	Code	Commit	Settings	Default clustering	Deviations from the tutorial
HiSTaR	deep graph VAE	PyTorch port of the official code (hicast/histar.py)	0039a3b	paper	3-start Gaussian mixture (scikit-learn, tied)	masking at inference removed in 'deterministic' variant only; headline uses published behaviour; per-section lambda_sim tuning described in the paper's supplement is NOT applied
GraphST	deep graph contrastive autoencoder	pip GraphST 1.1.1; tutorials from JinmiaoChenLab/GraphST	d62b0b7	tutorial	R mclust EEE on 20 PCs of the embedding	reported with and without the tutorial's label refinement (graphst vs graphst_refined)
BANKSY	neighbourhood-augmented features (non-deep)	pip pybanksy 1.3.5	9278996	tutorial	3-start Gaussian mixture (scikit-learn, tied)	both recommended lambdas are run: 0.2 (cell typing) and 0.8 (domain segmentation)
SpaGCN	graph convolutional network	pip SpaGCN 1.2.7	dc7a1c2	tutorial	iterative DEC clustering (own)	no histology image used (not available for all sections); reported with and without the hexagonal refinement
STAGATE	deep graph attention autoencoder	QIFEIDKN/STAGATE_pyG, installed from the clone	ae1158c	tutorial	R mclust EEE	radius expressed in spot spacings because our coordinates are on another scale (gives 5.8 neighbours per spot, as in the tutorial); torch_sparse replaced by a stub only if the compiled extension is unavailable (it was compiled from source here)
SEDR	deep masked graph autoencoder	JinmiaoChenLab/SEDR, installed from the clone	ef48360	tutorial	R mclust EEE (seed 2023)	mclust called as SEDR.mclust_R does, without its line that hard-codes the authors' R_HOME (breaks R's LAPACK in a running session)
SpaceFlow	deep graph infomax with spatial regularisation	pip SpaceFlow 1.0.4	58d1ab0	package default	Leiden at a fixed resolution	networkx>=3 alias for removed to_scipy_sparse_matrix; all-zero genes removed (cell_ranger HVG binning fails on them with pandas 3)
NichePCA	training-free (neighbour aggregation + PCA)	pip nichepca 0.1.0	package 0.1.0	package default	Leiden	
CCST	deep graph infomax (4 GCN layers)	xiaoyeye/CCST, re-implemented in scripts/new_methods.py bec	890de27	tutorial	k-means (n_init=100)	graph threshold chosen to give about 6 neighbours per cell (the authors hand-pick a threshold per dataset)
Spatial-MGCN	deep multi-view graph convolution	cs-wangbo/Spatial-MGCN, repository modules used directly	cf4412d	tutorial	k-means each epoch; epoch chosen by ARI against the annotation (LEAKAGE, see docs/LEAKAGE_AUDIT.md)	unannotated spots kept (the released script deletes them before training); label-free epoch rule
DeepST	deep graph + augmentation (no histology)	JiangBioLab/DeepST (deepstkit), used from the clone	e37997f	tutorial	Leiden at the given K + spatial refinement	no histology features; memory-heavy: run one job at a time
BASS	Bayesian hierarchical model	zhengli09/BASS, installed from the clone	5c2690f	tutorial	own (joint cell-type and domain model)	each section analysed separately (the authors' MERFISH/DLPFC analyses are multi-sample)
BayesSpace	Bayesian statistical model	edward130603/BayesSpace tag v1.5.1 (older release; newer one	ba8b422	package default	own MCMC	Visium only (needs a hexagonal spot lattice)
Linear diffusion	training-free			fixed rule	3-start Gaussian mixture (scikit-learn, tied)	settings fixed on DLPFC donor 1 before any other data were analysed
AnisoST	training-free, edge-preserving diffusion			fixed rule	3-start Gaussian mixture (scikit-learn, tied)	same as linear_diffusion
PCA	training-free			fixed rule	3-start Gaussian mixture (scikit-learn, tied)	

Supplementary Table S3. Mean ARI over sections and seeds per dataset with each method's default clustering and with matched clustering (the same 20-start best-likelihood Gaussian mixture on every embedding). The training-free baselines, HiSTaR and BANKSY use a 3-start Gaussian mixture as default clustering. SpaGCN, BASS and BayesSpace cluster internally and have the same value under both protocols.

Method	DLPFC default	DLPFC matched	STARmap default	STARmap matched	MERFISH default	MERFISH matched
Linear diffusion	0.543	0.558	0.708	0.705	0.481	0.497
AnisoST	0.526	0.551	0.564	0.622	0.377	0.363
PCA	0.388	0.409	0.270	0.294	0.151	0.130
HiSTaR	0.491	0.544	0.593	0.658	0.221	0.264
GraphST	0.514	0.531	0.388	0.484	0.251	0.254
BANKSY λ=0.8	0.494	0.518	0.740	0.694	0.428	0.410
BANKSY λ=0.2	0.514	0.518	0.502	0.529	0.211	0.222
SpaGCN	0.422	0.422	0.267	0.267	0.151	0.151
STAGATE	0.497	0.548	0.479	0.572	0.368	0.342
SEDR	0.515	0.551	0.593	0.236	0.406	0.167
SpaceFlow	0.312	0.326	0.628	0.668	0.389	0.260
NichePCA	0.321	0.506	0.692	0.604	0.475	0.403
CCST	0.437	0.359	0.505	0.435	0.212	0.157
Spatial-MGCN	0.418	0.391	0.303	0.245	0.182	0.133
DeepST	0.459	0.435	0.368	0.409	0.231	0.163
BASS	0.484	0.484	0.716	0.716	0.582	0.582
BayesSpace	0.437	0.437	–	–	–	–

Supplementary Table S4. As Supplementary Table S3 for normalised mutual information (NMI).

Method	DLPFC default	DLPFC matched	STARmap default	STARmap matched	MERFISH default	MERFISH matched
Linear diffusion	0.668	0.676	0.700	0.699	0.602	0.614
AnisoST	0.653	0.665	0.582	0.590	0.494	0.488
PCA	0.497	0.506	0.301	0.326	0.227	0.216
HiSTaR	0.637	0.660	0.602	0.653	0.375	0.405
GraphST	0.621	0.634	0.435	0.481	0.408	0.404
BANKSY λ=0.8	0.605	0.611	0.713	0.680	0.567	0.558
BANKSY λ=0.2	0.610	0.611	0.530	0.535	0.302	0.318
SpaGCN	0.555	0.555	0.317	0.317	0.238	0.238
STAGATE	0.641	0.668	0.492	0.552	0.490	0.473
SEDR	0.654	0.654	0.614	0.303	0.544	0.312
SpaceFlow	0.456	0.480	0.640	0.655	0.509	0.441
NichePCA	0.460	0.629	0.703	0.604	0.567	0.540
CCST	0.576	0.511	0.453	0.421	0.334	0.314
Spatial-MGCN	0.599	0.581	0.307	0.271	0.265	0.232
DeepST	0.628	0.572	0.489	0.429	0.370	0.261
BASS	0.639	0.639	0.690	0.690	0.638	0.638
BayesSpace	0.597	0.597	–	–	–	–

Supplementary Table S5. Difference in mean ARI between linear diffusion and each method, both with their default clustering; equal-weight mean of group means; 95% interval from a hierarchical bootstrap over five groups (three DLPFC donors, STARmap, MERFISH). Test only: the same estimate without the development sections. Verdict: baseline higher if the interval is above zero; unresolved otherwise; no comparison met the equivalence criterion (90% interval within ± 0.03).

Method	Δ	95% CI	DLPFC	STARmap	MERFISH	Test only	95% CI	Verdict
PCA	+0.247	(+0.141, +0.369)	+0.16	+0.44	+0.33	+0.272	(+0.158, +0.391)	baseline higher
SpaGCN	+0.227	(+0.113, +0.355)	+0.12	+0.44	+0.33	+0.257	(+0.135, +0.390)	baseline higher
Spatial-MGCN	+0.216	(+0.118, +0.329)	+0.12	+0.41	+0.30	+0.242	(+0.138, +0.369)	baseline higher
SpaceFlow	+0.174	(+0.093, +0.261)	+0.23	+0.08	+0.09	+0.183	(+0.089, +0.278)	baseline higher
DeepST	+0.169	(+0.060, +0.277)	+0.08	+0.34	+0.25	+0.202	(+0.073, +0.319)	baseline higher
CCST	+0.159	(+0.086, +0.231)	+0.11	+0.20	+0.27	+0.146	(+0.062, +0.236)	baseline higher
NichePCA	+0.138	(+0.031, +0.236)	+0.22	+0.02	+0.01	+0.105	(-0.036, +0.251)	baseline higher
GraphST	+0.128	(+0.015, +0.245)	+0.03	+0.32	+0.23	+0.136	(+0.010, +0.267)	baseline higher
BANKSY λ=0.2	+0.113	(+0.014, +0.219)	+0.03	+0.21	+0.27	+0.115	(+0.030, +0.215)	baseline higher
HiSTaR	+0.107	(+0.014, +0.203)	+0.05	+0.12	+0.26	+0.107	(-0.021, +0.228)	baseline higher
BayesSpace	+0.106	(+0.023, +0.198)	+0.11	–	–	+0.128	(+0.021, +0.231)	baseline higher
STAGATE	+0.096	(+0.012, +0.186)	+0.05	+0.23	+0.11	+0.106	(+0.030, +0.168)	baseline higher
AnisoST	+0.060	(+0.007, +0.119)	+0.02	+0.14	+0.10	+0.066	(+0.026, +0.105)	baseline higher
SEDR	+0.055	(+0.002, +0.117)	+0.03	+0.12	+0.08	+0.047	(-0.004, +0.101)	baseline higher
BANKSY λ=0.8	+0.034	(-0.015, +0.074)	+0.05	-0.03	+0.05	+0.030	(-0.045, +0.088)	unresolved
BASS	+0.014	(-0.055, +0.086)	+0.06	-0.01	-0.10	+0.004	(-0.081, +0.090)	unresolved

Supplementary Table S6. Primary analysis: as Supplementary Table S5 with matched clustering (the same 20-start best-likelihood Gaussian mixture on every embedding; SpaGCN, BASS and BayesSpace use their internal clustering).

Method	Δ	95% CI	DLPFC	STARmap	MERFISH	Test only	95% CI	Verdict
Spatial-MGCN	+0.265	(+0.137, +0.406)	+0.17	+0.46	+0.36	+0.309	(+0.187, +0.448)	baseline higher
PCA	+0.245	(+0.140, +0.359)	+0.15	+0.41	+0.37	+0.262	(+0.142, +0.385)	baseline higher
CCST	+0.242	(+0.142, +0.324)	+0.20	+0.27	+0.34	+0.250	(+0.126, +0.353)	baseline higher
SpaGCN	+0.238	(+0.124, +0.366)	+0.14	+0.44	+0.35	+0.269	(+0.141, +0.404)	baseline higher
DeepST	+0.200	(+0.102, +0.297)	+0.12	+0.30	+0.33	+0.202	(+0.088, +0.315)	baseline higher
SpaceFlow	+0.194	(+0.097, +0.269)	+0.23	+0.04	+0.24	+0.202	(+0.060, +0.289)	baseline higher
SEDR	+0.164	(-0.007, +0.352)	+0.01	+0.47	+0.33	+0.199	(-0.028, +0.415)	unresolved
BayesSpace	+0.121	(+0.037, +0.207)	+0.12	–	–	+0.141	(+0.035, +0.249)	baseline higher
BANKSY λ=0.2	+0.114	(+0.014, +0.225)	+0.04	+0.18	+0.28	+0.112	(+0.008, +0.230)	baseline higher
GraphST	+0.109	(+0.010, +0.209)	+0.03	+0.22	+0.24	+0.101	(-0.006, +0.210)	baseline higher
NichePCA	+0.070	(+0.017, +0.127)	+0.05	+0.10	+0.09	+0.066	(+0.014, +0.125)	baseline higher
HiSTaR	+0.064	(-0.018, +0.163)	+0.01	+0.05	+0.23	+0.066	(-0.056, +0.197)	unresolved
STAGATE	+0.064	(+0.000, +0.133)	+0.01	+0.13	+0.15	+0.073	(+0.014, +0.137)	baseline higher
AnisoST	+0.048	(-0.007, +0.105)	+0.01	+0.08	+0.13	+0.058	(+0.005, +0.127)	unresolved
BANKSY λ=0.8	+0.044	(-0.007, +0.095)	+0.04	+0.01	+0.09	+0.054	(-0.009, +0.117)	unresolved
BASS	+0.025	(-0.044, +0.101)	+0.07	-0.01	-0.09	+0.015	(-0.072, +0.109)	unresolved

Supplementary Table S7. Rank of each method in the hierarchical bootstrap with matched clustering (first three columns) and with default clustering (last three); methods with all 20 sections.

Method	P(rank 1)	Median rank	95% interval	P(rank 1), default	Median rank, default	95% interval, default
Linear diffusion	0.60	1	1–3	0.56	1	1–3
BASS	0.25	2	1–10	0.35	2	1–9
BANKSY λ=0.8	0.02	3	2–9	0.06	3	1–9
AnisoST	0.03	4	1–7	0.00	5	2–7
HiSTaR	0.06	6	1–8	0.00	8	3–11
STAGATE	0.01	6	2–8	0.01	7	3–11
NichePCA	0.00	6	3–10	0.00	10	4–16
GraphST	0.00	8	4–10	0.01	9	3–12
BANKSY λ=0.2	0.00	9	4–10	0.01	8	3–12
SEDR	0.03	10	1–15	0.01	4	2–6
DeepST	0.00	11	11–14	0.00	12	8–13
SpaceFlow	0.00	12	7–16	0.00	12	6–16
SpaGCN	0.00	14	11–16	0.00	15	11–16
CCST	0.00	14	11–16	0.00	11	8–14
PCA	0.00	14	11–16	0.00	16	13–16
Spatial-MGCN	0.00	15	12–16	0.00	14	11–15

Supplementary Table S8. Seed-to-seed standard deviation of ARI (mean over sections; default clustering). Methods run with one seed have no value.

Method	DLPFC	STARmap	MERFISH	Seeds (D/S/M)
Linear diffusion	0.039	0.012	0.054	3/3/3
AnisoST	0.022	0.042	0.015	3/3/3
PCA	0.015	0.019	0.015	3/3/3
HiSTaR	0.047	0.180	0.053	3/3/3
GraphST	–	0.086	0.031	1/3/3
BANKSY λ=0.8	0.067	0.006	0.041	3/3/3
BANKSY λ=0.2	0.056	0.063	0.022	3/3/3
SpaGCN	0.038	0.030	0.016	3/3/3
STAGATE	–	–	–	1/1/1
SEDR	0.042	0.083	0.060	3/3/3
SpaceFlow	0.032	0.101	0.030	3/3/3
NichePCA	0.023	0.034	0.029	3/3/3
CCST	–	–	–	1/1/1
Spatial-MGCN	–	–	–	1/1/1
DeepST	0.042	0.030	0.028	3/3/3
BASS	–	–	–	1/1/1
BayesSpace	–	–	–	1/0/0

Supplementary Table S9. Initialisation experiment: one fixed embedding per section refitted 20 times with a single-start Gaussian mixture (20 sections per method). Range and SD of ARI across the 20 fits; best-by-ARI minus mean: ARI gain of choosing the best fit by agreement with the labels; best-likelihood minus mean: gain of choosing the fit with the highest likelihood; ρ: median Spearman correlation between log-likelihood and ARI.

Method	Median range	IQR	Mean SD	Best-by-ARI minus mean	Best-likelihood minus mean	ρ DLPFC	ρ STARmap	ρ MERFISH
SpaceFlow	0.07	0.05–0.11	0.028	0.036	+0.005	-0.05	+0.60	+0.51
GraphST	0.07	0.02–0.14	0.031	0.044	+0.017	+0.36	+0.23	-0.04
Spatial-MGCN	0.12	0.08–0.15	0.033	0.048	-0.017	+0.07	+0.12	+0.19
DeepST	0.12	0.08–0.24	0.046	0.063	-0.016	+0.14	+0.38	-0.26
PCA	0.15	0.08–0.22	0.039	0.055	+0.020	-0.16	+0.17	-0.25
SEDR	0.17	0.10–0.24	0.060	0.099	+0.038	+0.50	-0.64	-0.20
CCST	0.17	0.08–0.29	0.060	0.076	-0.062	-0.04	+0.26	-0.17
AnisoST	0.17	0.14–0.26	0.060	0.091	+0.039	+0.48	+0.47	-0.36
STAGATE	0.18	0.12–0.23	0.053	0.070	+0.041	+0.62	+0.57	-0.22
HiSTaR	0.19	0.14–0.25	0.068	0.113	+0.069	+0.22	+0.53	-0.57
Linear diffusion	0.19	0.16–0.26	0.064	0.078	+0.060	+0.88	+0.26	-0.01
NichePCA	0.21	0.16–0.28	0.067	0.095	+0.037	+0.65	+0.54	+0.22
BANKSY λ=0.2	0.22	0.13–0.31	0.064	0.098	+0.043	+0.25	+0.57	+0.10
BANKSY λ=0.8	0.22	0.18–0.29	0.080	0.100	+0.008	+0.19	+0.76	-0.08

Supplementary Table S10. R mclust 6.0.1 (EEE) on fixed embeddings. Seed range: ARI range over ten R random seeds with the default initialisation, which uses a random subset of 2000 observations when there are more (sections with more than 2000 spots or cells only; STARmap sections are smaller and do not vary). Noise range: ARI range over five repetitions with 1% Gaussian noise when EM is initialised on all observations, which is deterministic. Full minus seeded mean: ARI of the full-data initialisation minus the mean of the ten seeded fits.

Method	Embeddings	Seed range (mean)	Seed range (max)	Range $>$0.1	Noise range, full init.	Full minus seeded mean
Linear diffusion	20	0.19	0.35	88%	0.14	-0.001
NichePCA	20	0.18	0.42	82%	0.18	+0.004
PCA	20	0.16	0.46	65%	0.12	-0.023
BANKSY λ=0.8	20	0.15	0.34	65%	0.11	-0.005
SEDR	20	0.14	0.33	71%	0.08	+0.003
STAGATE	20	0.13	0.27	65%	0.09	+0.019
HiSTaR	20	0.12	0.32	59%	0.09	-0.004
AnisoST	20	0.11	0.22	53%	0.06	-0.034
Spatial-MGCN	20	0.10	0.40	41%	0.06	+0.011
BANKSY λ=0.2	20	0.10	0.28	53%	0.07	-0.004
SpaceFlow	20	0.10	0.20	47%	0.08	-0.001
CCST	20	0.09	0.35	24%	0.07	+0.012
GraphST	20	0.09	0.26	41%	0.07	-0.006
DeepST	20	0.06	0.17	24%	0.03	+0.009

Supplementary Table S11. Change in mean ARI when each method's default clustering is replaced by matched clustering (20-start best-likelihood Gaussian mixture) on the same embedding (20 sections). The two protocols can differ in algorithm and number of starts.

Method	Mean change in ARI	Sections improved	Wilcoxon p (unadjusted)
NichePCA	+0.080	13/20	0.040
HiSTaR	+0.053	14/20	0.004
STAGATE	+0.038	12/20	0.202
GraphST	+0.025	13/20	0.114
AnisoST	+0.020	14/20	0.020
Linear diffusion	+0.012	11/20	0.102
PCA	+0.011	7/20	0.368
BANKSY λ=0.2	+0.009	12/20	0.177
BANKSY λ=0.8	+0.003	9/20	0.679
SpaceFlow	-0.018	7/20	0.368
DeepST	-0.025	6/20	0.312
Spatial-MGCN	-0.037	6/20	0.033
CCST	-0.071	4/20	0.008
SEDR	-0.092	6/20	0.070

Supplementary Table S12. Metric panel from the standardised runner (shared preprocessing for all methods; mean over the three datasets of the per-dataset means). Values differ slightly from Supplementary Table S3 because the preprocessing is shared. Missed domains: annotated domains with no matched cluster; small-domain recall: recall of domains below 5% of spots; CHAOS and PAS: spatial fragmentation (lower is better). Runs: 60 per method (20 sections, 3 seeds), except DeepST (20, seed 0) and BayesSpace (12 Visium sections, seed 0).

Method	ARI	NMI	Macro-F1	Missed domains	Small-domain recall	CHAOS↓	PAS↓	Boundary F1	Runs
BASS	0.545	0.650	0.613	2.07	0.648	1.063	0.037	0.667	65
Linear diffusion	0.525	0.652	0.605	2.28	0.674	1.032	0.009	0.625	64
BANKSY λ=0.8	0.519	0.629	0.590	2.12	0.650	1.040	0.027	0.628	66
BayesSpace	0.500	0.638	0.507	2.50	0.565	1.045	0.054	0.655	13
AnisoST	0.489	0.599	0.557	2.57	0.580	1.044	0.044	0.629	66
NichePCA	0.468	0.603	0.568	2.10	0.513	1.037	0.011	0.564	63
STAGATE	0.446	0.582	0.584	1.75	0.721	1.098	0.064	0.620	64
SEDR	0.444	0.596	0.551	2.31	0.526	1.056	0.038	0.627	64
SpaceFlow	0.432	0.566	0.531	2.10	0.452	1.054	0.027	0.575	64
BANKSY λ=0.2	0.432	0.539	0.523	2.65	0.589	1.237	0.161	0.602	66
HiSTaR	0.416	0.565	0.556	2.34	0.613	1.126	0.085	0.612	64
GraphST	0.406	0.537	0.540	2.02	0.423	1.114	0.106	0.620	63
CCST	0.391	0.507	0.448	3.54	0.403	1.081	0.045	0.552	65
DeepST	0.379	0.545	0.551	1.63	0.695	1.368	0.187	0.577	23
PCA	0.357	0.452	0.470	2.96	0.533	1.354	0.275	0.608	64
SpaGCN	0.353	0.462	0.471	2.43	0.568	1.294	0.266	0.615	64
Spatial-MGCN	0.342	0.471	0.470	2.25	0.618	1.314	0.213	0.563	62

Supplementary Table S13. Run time, peak memory and estimated energy of one whole job (data loading, embedding and clustering) in the standardised runner on one CPU thread. Energy is estimated at 10 W per busy core.

Method	Median time per job (s)	Peak memory (GB)	Energy per job (Wh, estimate)
BANKSY λ=0.8	23	4.8	0.06
BANKSY λ=0.2	23	4.8	0.06
AnisoST	24	3.7	0.06
Linear diffusion	24	3.7	0.06
PCA	26	3.5	0.07
NichePCA	57	9.1	0.15
SpaGCN	57	8.5	0.19
SEDR	105	9.4	0.28
HiSTaR	120	3.5	0.35
BASS	732	10.1	1.89
DeepST	791	10.8	2.56
SpaceFlow	908	3.1	2.66
STAGATE	1020	3.3	2.69
GraphST	1254	4.9	3.34
Spatial-MGCN	1512	7.9	3.67
BayesSpace	1892	12.5	5.49
CCST	2564	5.5	9.47

Supplementary Table S14. Mean ARI with each method's default clustering in its published pipeline (used in the main text) and in the standardised runner (shared preprocessing, used for Supplementary Tables S12 and S13), and their difference (runner minus pipeline).

Method	DLPFC pipeline	DLPFC runner	DLPFC diff.	STARmap pipeline	STARmap runner	STARmap diff.	MERFISH pipeline	MERFISH runner	MERFISH diff.
Linear diffusion	0.543	0.569	+0.026	0.708	0.733	+0.025	0.481	0.497	+0.016
AnisoST	0.526	0.554	+0.028	0.564	0.621	+0.057	0.377	0.363	-0.013
PCA	0.388	0.417	+0.029	0.270	0.295	+0.025	0.151	0.130	-0.021
HiSTaR	0.491	0.497	+0.006	0.593	0.515	-0.077	0.221	0.237	+0.016
GraphST	0.514	0.502	-0.013	0.388	0.388	+0.000	0.251	0.260	+0.009
BANKSY λ=0.8	0.494	0.494	+0.000	0.740	0.740	+0.000	0.428	0.428	+0.000
BANKSY λ=0.2	0.514	0.514	-0.000	0.502	0.502	+0.000	0.211	0.211	+0.000
SpaGCN	0.422	0.422	+0.000	0.267	0.259	-0.008	0.151	0.113	-0.039
STAGATE	0.497	0.509	+0.011	0.479	0.550	+0.071	0.368	0.331	-0.037
SEDR	0.515	0.515	+0.000	0.593	0.593	+0.000	0.406	0.406	+0.000
SpaceFlow	0.312	0.312	+0.000	0.628	0.628	+0.000	0.389	0.389	+0.000
NichePCA	0.321	0.321	+0.000	0.692	0.692	+0.000	0.475	0.475	+0.000
CCST	0.437	0.438	+0.001	0.505	0.498	-0.007	0.212	0.222	+0.010
Spatial-MGCN	0.418	0.414	-0.004	0.303	0.304	+0.001	0.182	0.175	-0.006
DeepST	0.459	0.456	-0.003	0.368	0.343	-0.025	0.231	0.220	-0.011
BASS	0.484	0.490	+0.005	0.716	0.718	+0.001	0.582	0.561	-0.022
BayesSpace	0.437	0.437	+0.000	–	–	–	–	–	–

Supplementary Table S15. Sensitivity to the number of domains: mean ARI over the 20 sections when the seed-0 embedding is clustered with matched clustering (20-start tied Gaussian mixture) for K from two below to two above the annotated number, and for the K in that range with the lowest BIC (label-free choice; it was almost always the largest K allowed).

Method	$K-2$	$K-1$	True K	$K+1$	$K+2$	BIC choice
Linear diffusion	0.487	0.570	0.579	0.519	0.477	0.477
AnisoST	0.418	0.509	0.528	0.492	0.446	0.446
BANKSY λ=0.8	0.409	0.459	0.511	0.535	0.508	0.508
STAGATE	0.421	0.481	0.500	0.480	0.432	0.441
HiSTaR	0.356	0.453	0.496	0.467	0.423	0.423
NichePCA	0.438	0.489	0.487	0.505	0.480	0.480
BANKSY λ=0.2	0.264	0.406	0.468	0.460	0.441	0.441
GraphST	0.375	0.455	0.452	0.408	0.397	0.395
SEDR	0.326	0.374	0.421	0.399	0.384	0.384
SpaceFlow	0.324	0.381	0.361	0.330	0.328	0.328
DeepST	0.259	0.346	0.357	0.386	0.382	0.381
PCA	0.270	0.283	0.328	0.316	0.314	0.316
CCST	0.190	0.265	0.320	0.362	0.377	0.377
Spatial-MGCN	0.272	0.298	0.304	0.316	0.310	0.313

Supplementary Table S16. Reproduction of HiSTaR on the twelve DLPFC sections.

Setting	Median ARI	Mean ARI	95% CI of mean	Source
Reported (HiSTaR article)	0.65 (median)	–	–	HiSTaR article
Official code, 3-start GMM (3 seeds)	0.509	0.491	0.454–0.525	this study
Masking at inference removed	0.502	0.487	0.441–0.529	this study
R mclust as in the official pipeline (seed 0)	–	0.449	–	this study

Supplementary Table S17. Mean ARI of the final training epoch (used in the benchmark) and of the epoch with the highest ARI against the annotation, as selected by the released test scripts (seed 0).

Dataset	Sections	Final epoch	Label-selected epoch	Difference
DLPFC	12	0.418	0.551	+0.133
STARmap	3	0.303	0.687	+0.384
MERFISH	5	0.182	0.221	+0.039
All	20	0.342	0.489	+0.147

Supplementary Table S18. Automated screening of released code for patterns of label use (number of candidate lines per category; L1 result chosen by agreement with labels, L2 number of domains from labels, L3 unannotated spots removed, L4 settings keyed by section, L5 loops combined with a label metric). Candidate lines are not findings: each was read manually; the 14 repositories are the benchmarked methods with released code plus SPARK, SpatialPCA and conST, which were screened but not benchmarked, and only the label-selected epoch in Spatial-MGCN's test scripts was confirmed as L1 (Supplementary Note 5).

Repository	L1	L2	L3	L4	L5
BASS	0	0	0	0	0
Banksy_py	2	0	4	0	3
BayesSpace	0	0	2	0	0
CCST	0	5	0	0	1
DeepST	0	1	3	3	1
GraphST	0	0	0	0	0
SEDR	0	0	2	1	0
SPARK	0	0	7	0	0
STAGATE_pyG	0	0	0	0	0
SpaGCN	0	2	0	0	0
SpaceFlow	0	0	0	0	0
Spatial-MGCN	10	3	5	0	2
SpatialPCA	0	0	2	0	0
conST	0	0	2	0	0

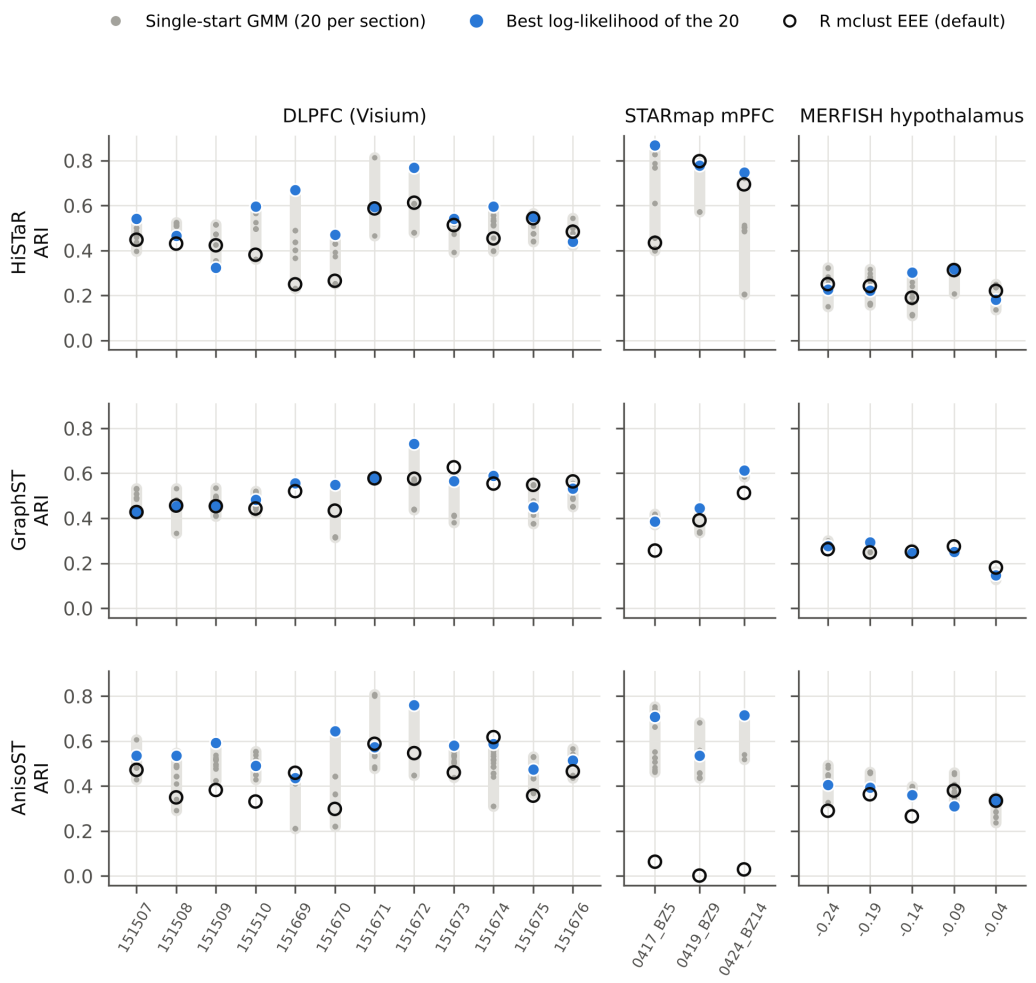
Supplementary Table S19. Pilot invariance test (three training-free methods, two sections): agreement (ARI) between the partition on the original data and on transformed data that should not change the answer. The methods are invariant to rescaling, rotating and reflecting the coordinates and to rescaling expression, but not fully to spot order, because the Gaussian mixture initialisation depends on it.

Section and method	permute spots	rescale coordinates x100	rescale coordinates x0.01	translate coordinates	rotate coordinates 37°	reflect coordinates	permute features	scale expression x2
DLPFC 151673 PCA	0.99	1.00	1.00	1.00	1.00	1.00	1.00	1.00
DLPFC 151673 Linear diffusion	0.97	1.00	1.00	1.00	1.00	1.00	1.00	1.00
DLPFC 151673 AnisoST	0.89	1.00	1.00	1.00	1.00	1.00	1.00	1.00
MERFISH -0.14 PCA	0.39	1.00	1.00	1.00	1.00	1.00	1.00	1.00
MERFISH -0.14 Linear diffusion	0.94	1.00	1.00	1.00	1.00	1.00	1.00	1.00
MERFISH -0.14 AnisoST	0.94	1.00	1.00	1.00	1.00	1.00	1.00	1.00

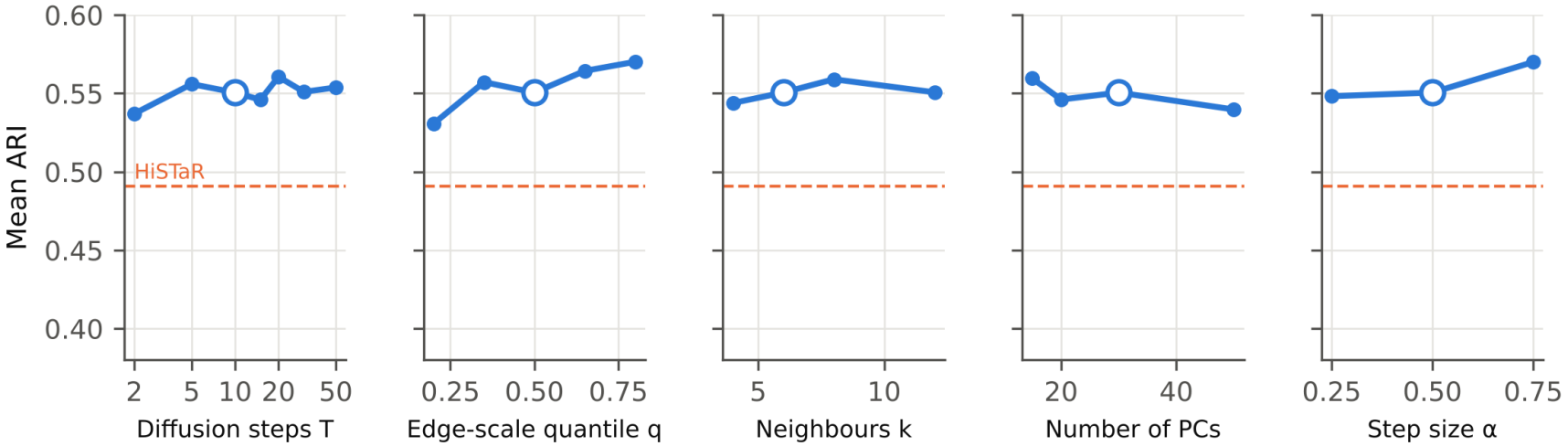
Supplementary Table S20. Which method was run in which experiment (sections completed / expected; seeds). n/a: the experiment needs an embedding and the method clusters internally.

Method	ARI, paper pipeline (sections / seeds)	Standardised runner + metric panel + resources (sections / seeds)	Lottery (20 starts)	R mclust perturbation	Invariance tests (sections)
Linear diffusion	2not run / 3	2not run / 3	2not run	2not run	2/20
AnisoST	2not run / 3	2not run / 3	2not run	2not run	2/20
PCA	2not run / 3	2not run / 3	2not run	2not run	2/20
BANKSY λ=0.2	2not run / 3	2not run / 3	2not run	2not run	not run
BANKSY λ=0.8	2not run / 3	2not run / 3	2not run	2not run	not run
HiSTaR	2not run / 3	2not run / 3	2not run	2not run	not run
GraphST	2not run / 3	2not run / 3	2not run	2not run	not run
SpaGCN	2not run / 3	2not run / 3	n/a (no embedding)	n/a (no embedding)	not run
STAGATE	2not run / 1	2not run / 3	2not run	2not run	not run
SEDR	2not run / 3	2not run / 3	2not run	2not run	not run
SpaceFlow	2not run / 3	2not run / 3	2not run	2not run	not run
NichePCA	2not run / 3	2not run / 3	2not run	2not run	not run
CCST	2not run / 1	2not run / 3	2not run	2not run	not run
Spatial-MGCN	2not run / 1	2not run / 3	2not run	2not run	not run
DeepST	2not run / 3	2not run / 1	2not run	2not run	not run
BASS	2not run / 1	2not run / 3	n/a (no embedding)	n/a (no embedding)	not run
BayesSpace	12/12 / 1	12/12 / 1	n/a (no embedding)	n/a (no embedding)	not run

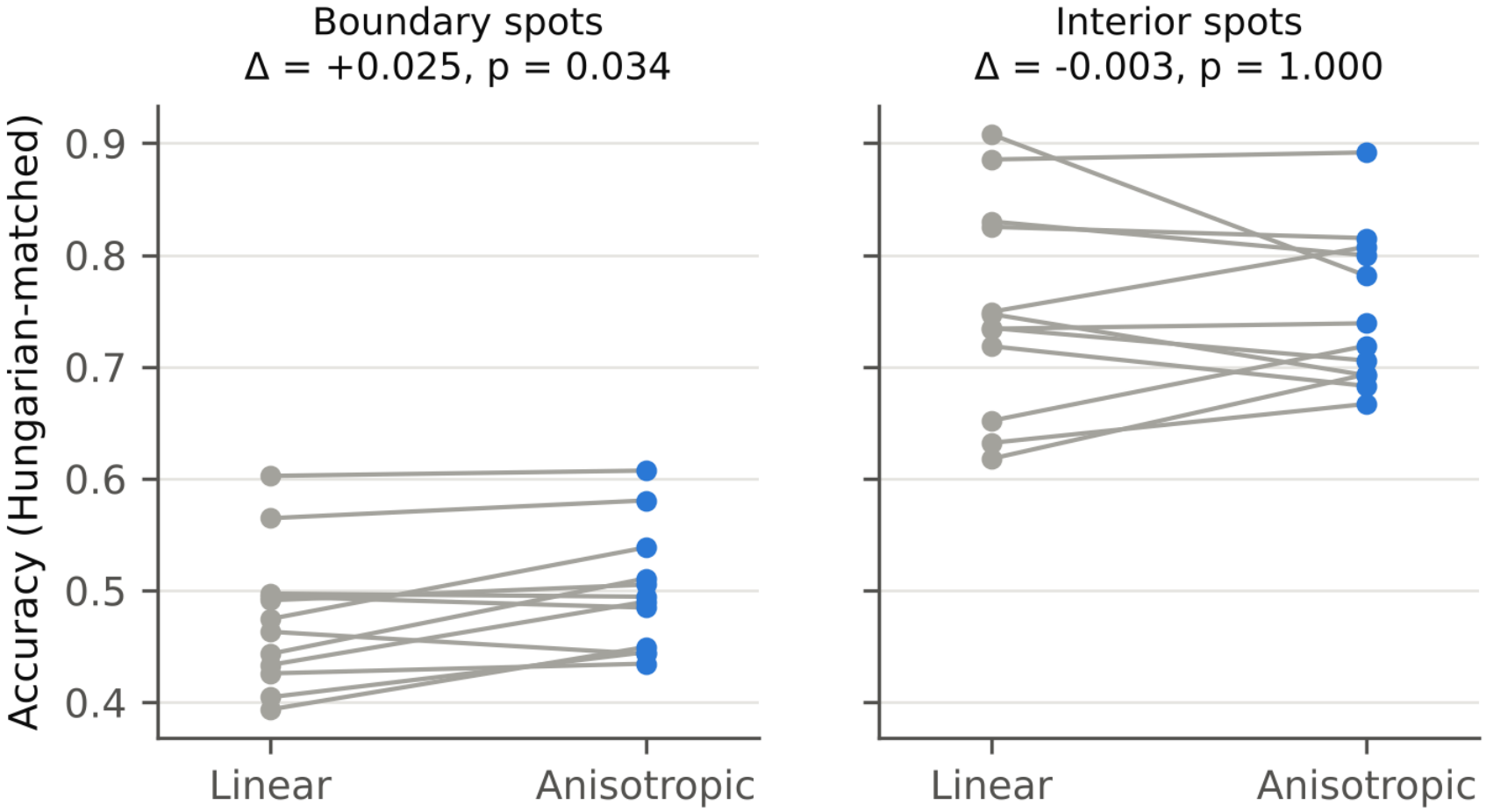
Supplementary Figures



Supplementary Figure S1. Initialisation experiment for three representative methods (HiSTaR, GraphST and AnisoST), per section: ARI of 20 single-start fits of one fixed embedding (grey), the best-likelihood fit (blue) and R mclust with default settings (open circle).



Supplementary Figure S2. Sensitivity of linear diffusion (diffusion steps T, neighbours k, number of PCs, step size α) and of AnisoST (edge-scale quantile q) on the 12 DLPFC sections, one hyper-parameter at a time (open circle: default; dashed line: HiSTaR mean ARI).



Supplementary Figure S3. Accuracy on spots at layer boundaries and in layer interiors on the 12 DLPFC sections, linear diffusion versus AnisoST (labelled Anisotropic); one line per section; Wilcoxon signed-rank test, $n = 12$ sections.